

# Synoptic CB: Porewater Nutrients

2022-2025 Samples

2026-05-07

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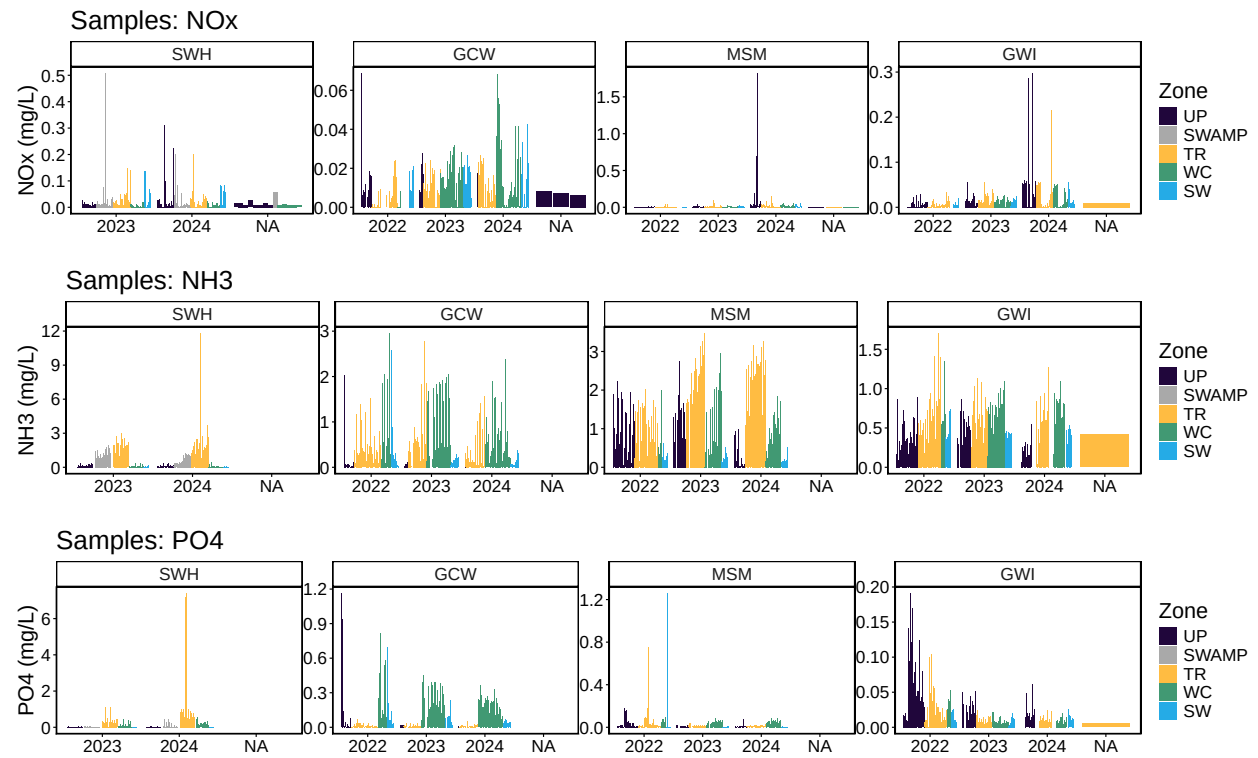
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### 0.1 Grab 2022 and 2023 All Data Files

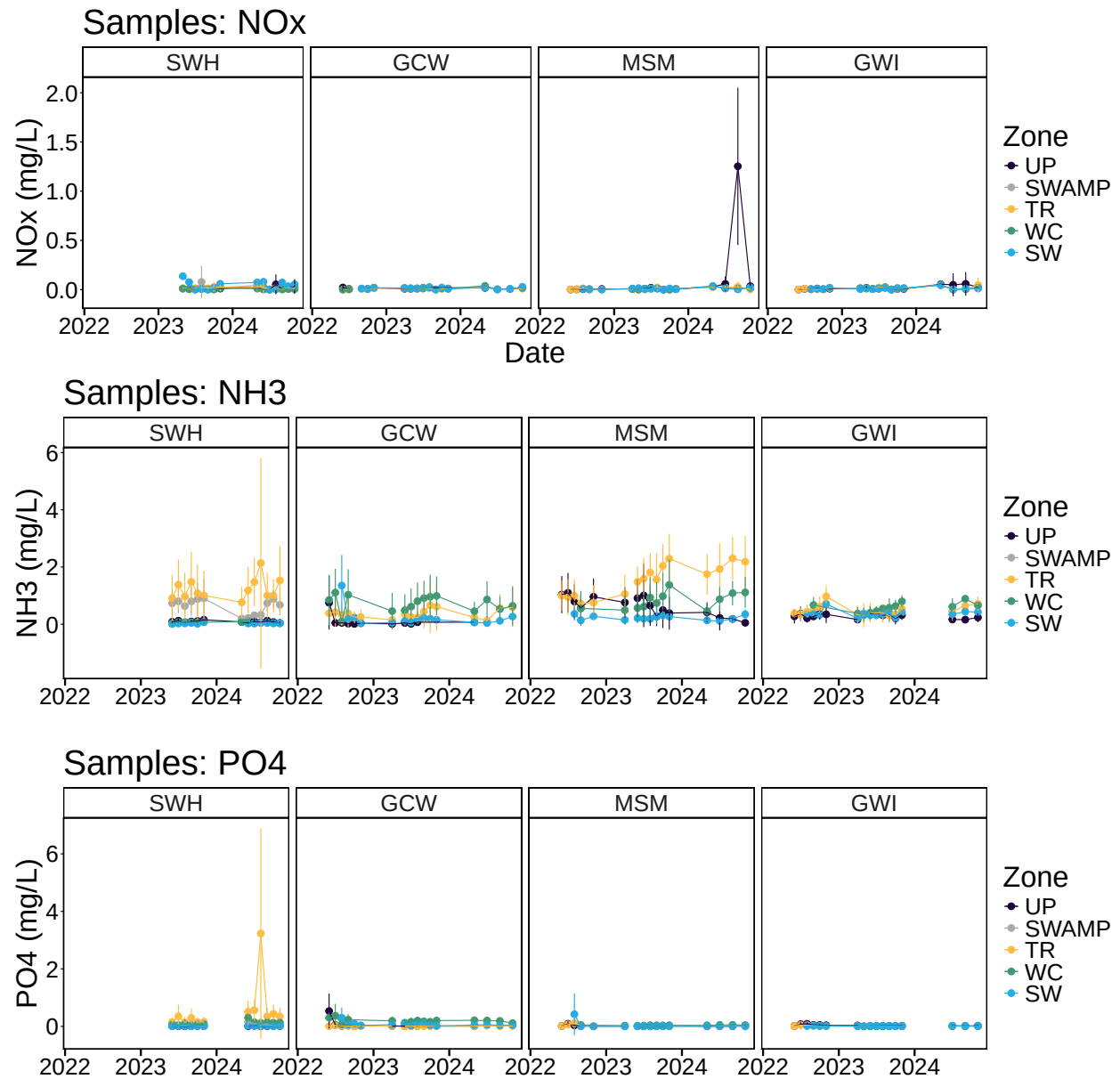
##Make Relevant Metadata Sheet

## 0.2 Write out files

## 0.3 Visualize Data by Plot

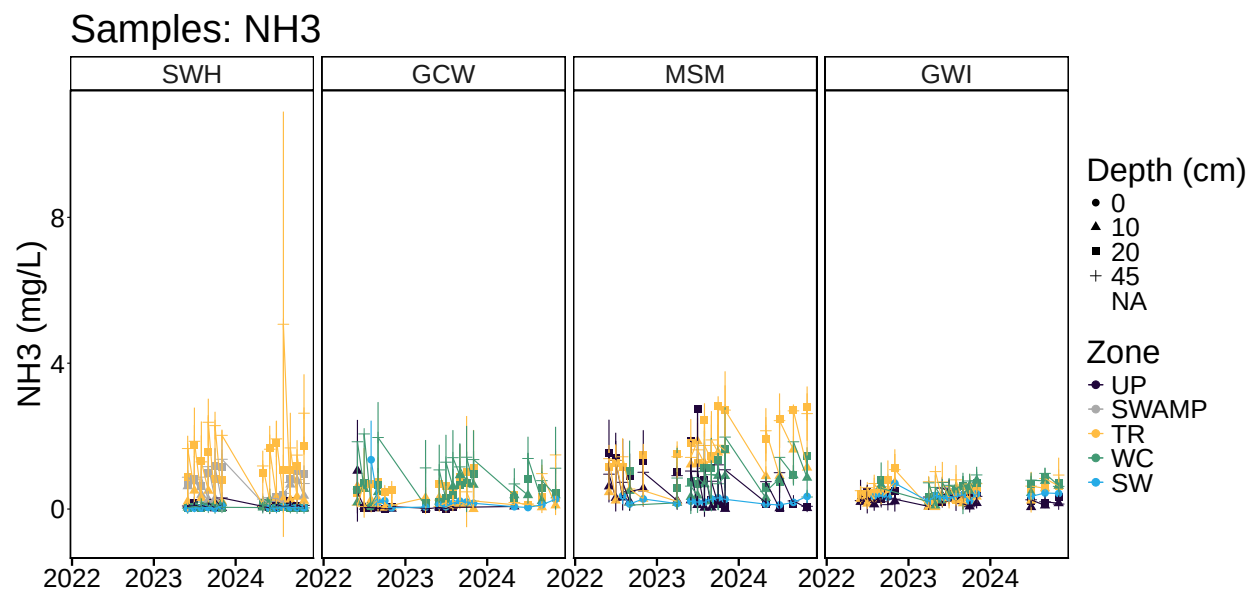
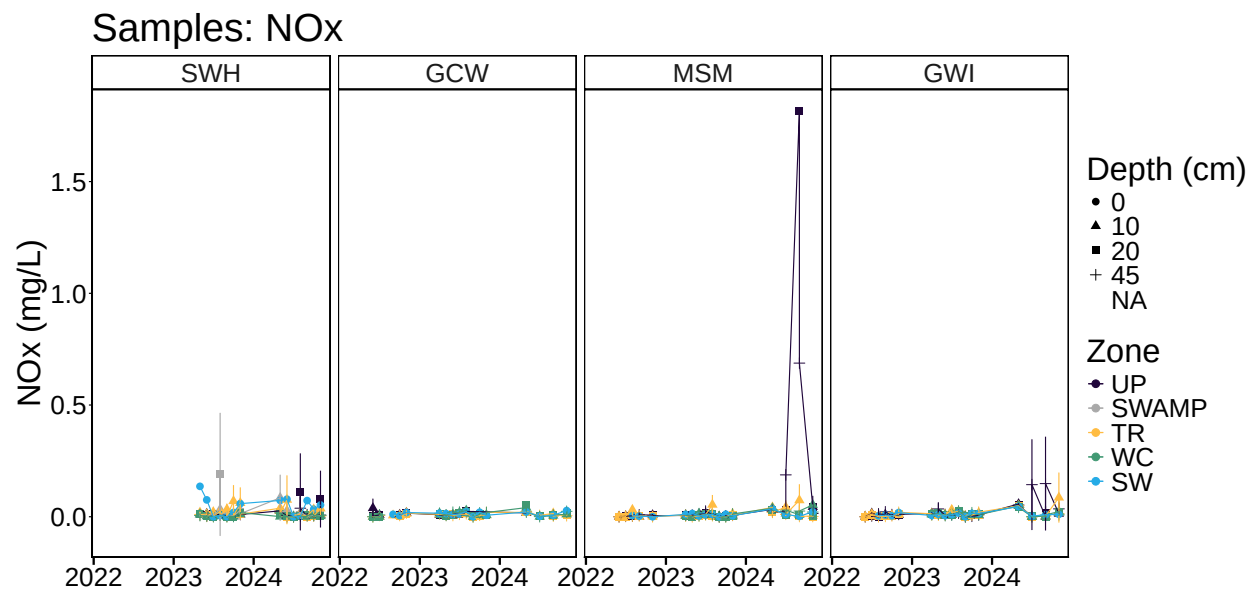


#### 0.4 Summarized data for Site and Zone

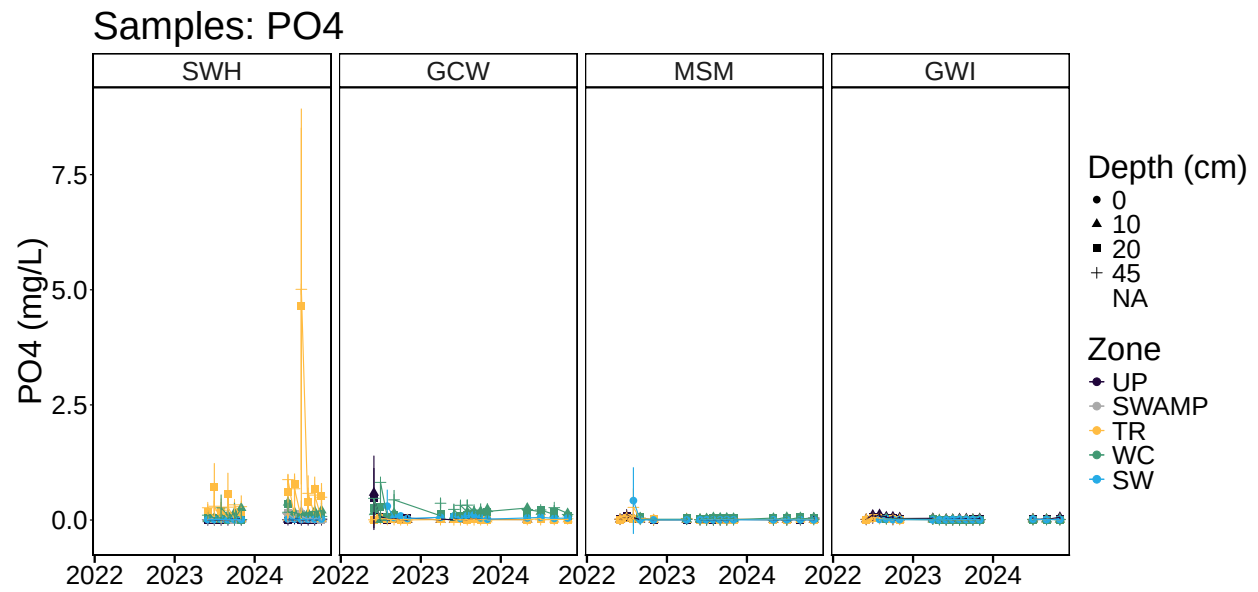


#### 0.5 Summarized data for Depth, Site and Zone

## \$'1'

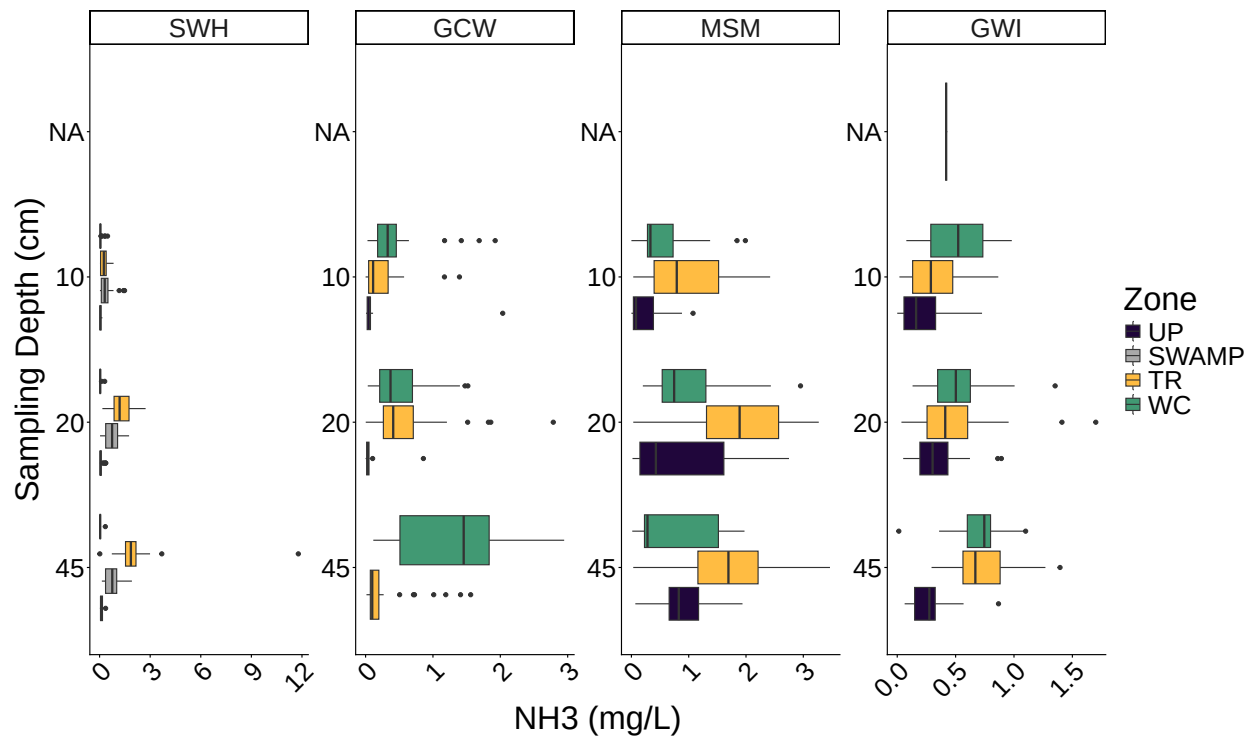
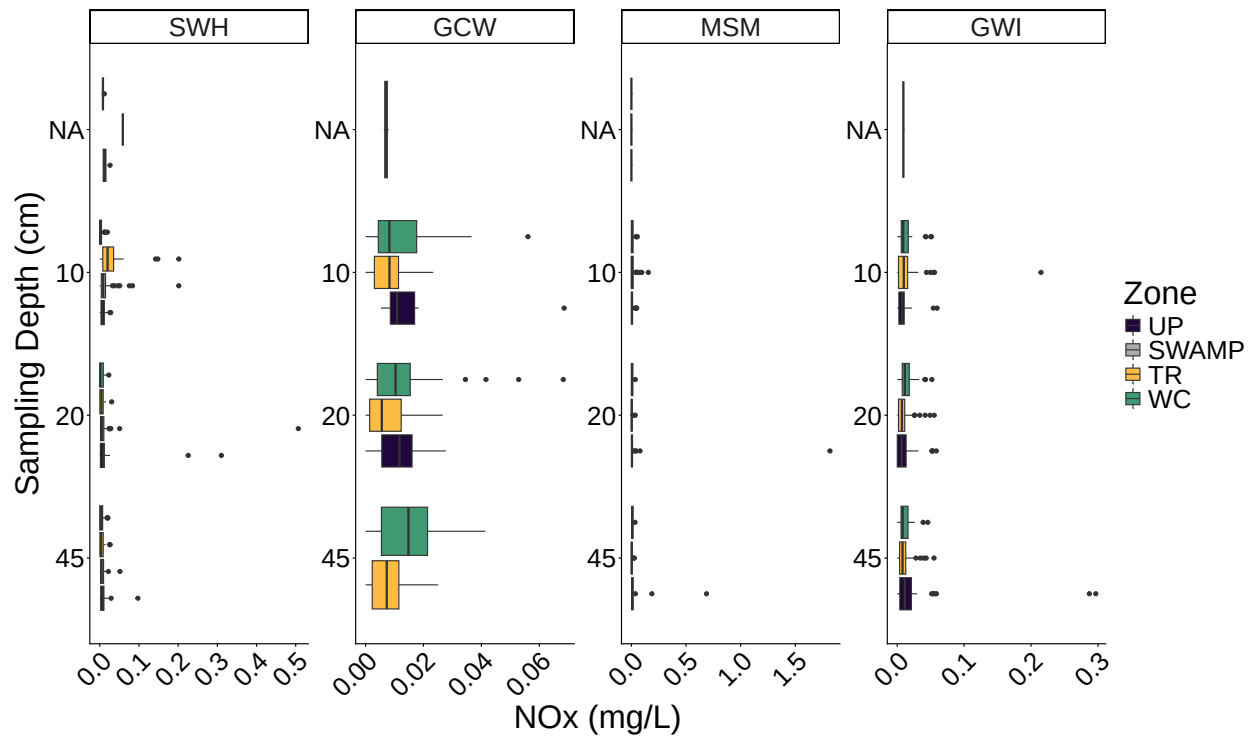


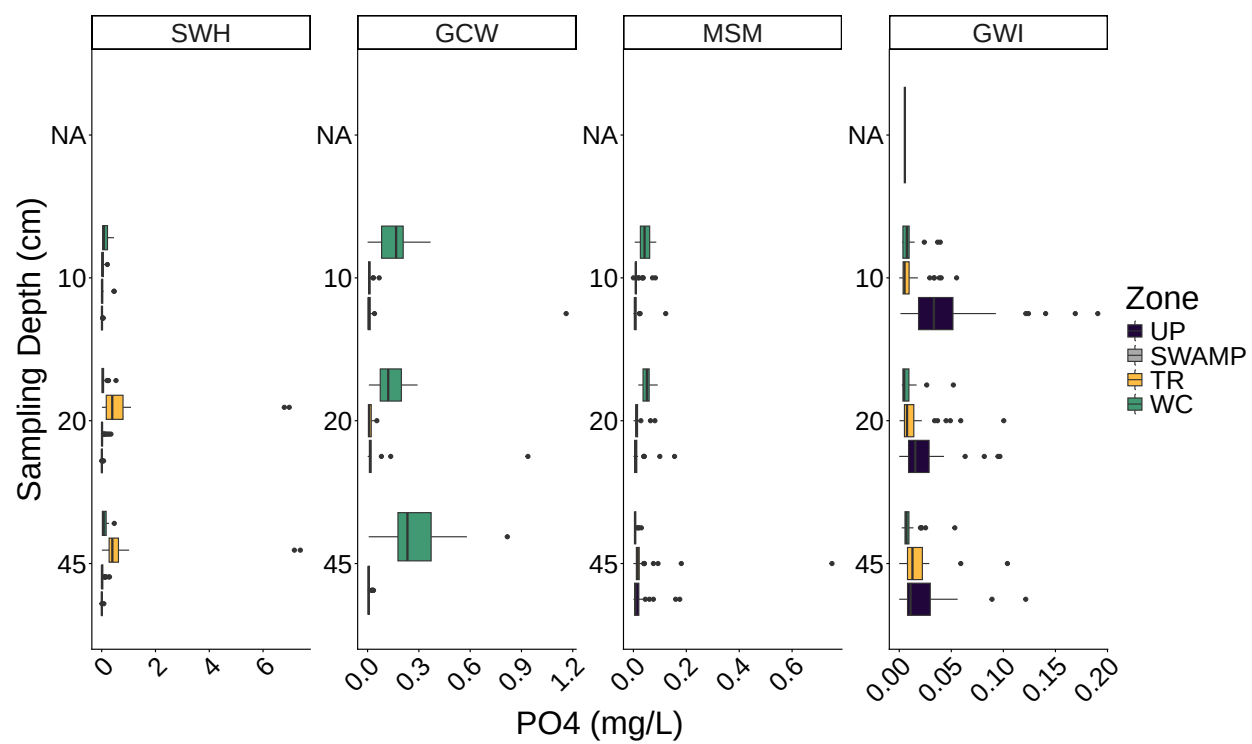
##  
## \$'2'



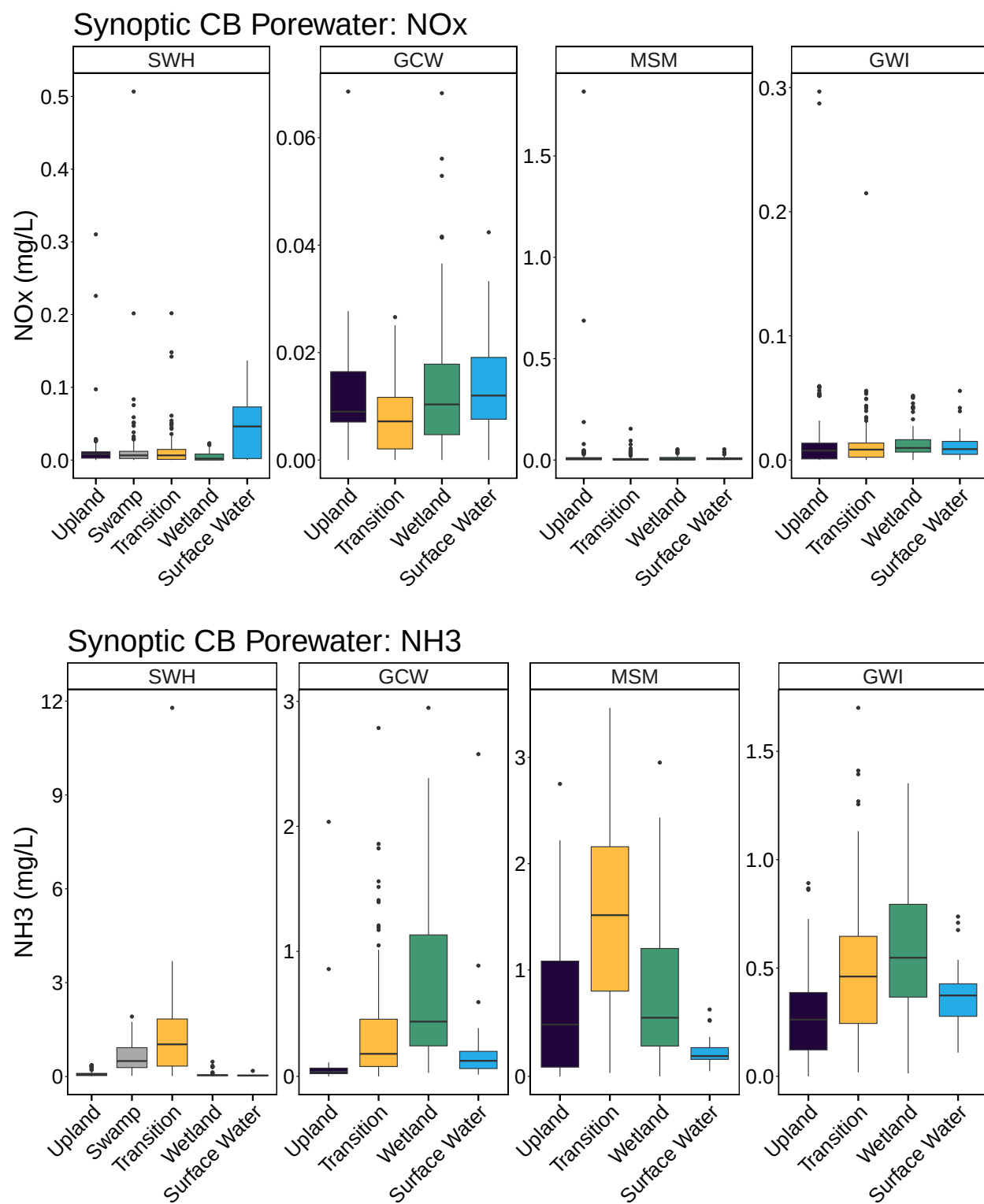
```
##
## attr("class")
## [1] "list"      "ggarrange"
```

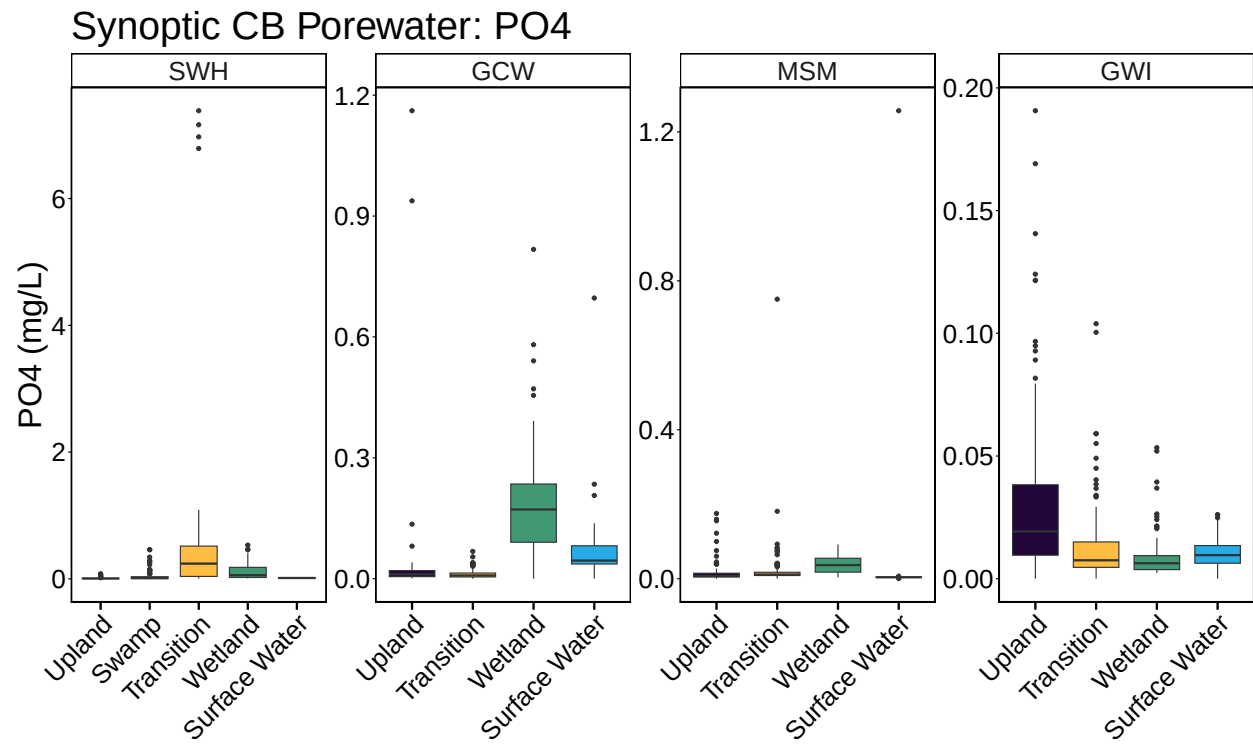
## 0.6 Boxplot by depth





## 0.7 Boxplot summary





```
## pdf
## 2
```

```
## pdf
## 2
```

```
## pdf
## 2
```

## 0.8 Calculate summary metrics

```
##Calculate number of samples
```

Table 1: NOx Summary Statistics

| Site | Zone  | min_NOx_mgL | max_NOx_mgL | mean_NOx_mgL | sd_NOx_mgL |
|------|-------|-------------|-------------|--------------|------------|
| SWH  | UP    | 0           | 0.3103      | 0.0133424    | 0.0356850  |
| SWH  | SWAMP | 0           | 0.5067      | 0.0158880    | 0.0496029  |
| SWH  | TR    | 0           | 0.2018      | 0.0141310    | 0.0272109  |
| SWH  | WC    | 0           | 0.0231      | 0.0047425    | 0.0059843  |
| SWH  | SW    | 0           | 0.1369      | 0.0412610    | 0.0383366  |
| GCW  | UP    | 0           | 0.0686      | 0.0128767    | 0.0123700  |
| GCW  | TR    | 0           | 0.0266      | 0.0080569    | 0.0071370  |
| GCW  | WC    | 0           | 0.0683      | 0.0131272    | 0.0125063  |
| GCW  | SW    | 0           | 0.0424      | 0.0129289    | 0.0092322  |
| MSM  | UP    | 0           | 1.8177      | 0.0341654    | 0.1871449  |
| MSM  | TR    | 0           | 0.1545      | 0.0081792    | 0.0184099  |
| MSM  | WC    | 0           | 0.0532      | 0.0094222    | 0.0115717  |
| MSM  | SW    | 0           | 0.0532      | 0.0093111    | 0.0113485  |
| GWI  | UP    | 0           | 0.2968      | 0.0162099    | 0.0388579  |
| GWI  | TR    | 0           | 0.2149      | 0.0127566    | 0.0214213  |
| GWI  | WC    | 0           | 0.0518      | 0.0134500    | 0.0127942  |
| GWI  | SW    | 0           | 0.0557      | 0.0111958    | 0.0110529  |

Table 2: NH3 Summary Statistics

| Site | Zone  | min_NH3_mgL | max_NH3_mgL | mean_NH3_mgL | sd_NH3_mgL |
|------|-------|-------------|-------------|--------------|------------|
| SWH  | UP    | 0.0112      | 0.3640      | 0.0901473    | 0.0845119  |
| SWH  | SWAMP | 0.0221      | 1.9169      | 0.6283730    | 0.4426947  |
| SWH  | TR    | 0.0141      | 11.7845     | 1.2230698    | 1.2989187  |
| SWH  | WC    | 0.0120      | 0.4729      | 0.0594184    | 0.0717974  |
| SWH  | SW    | 0.0064      | 0.1816      | 0.0354944    | 0.0295265  |
| GCW  | UP    | 0.0000      | 2.0360      | 0.1315903    | 0.3837326  |
| GCW  | TR    | 0.0000      | 2.7866      | 0.3628960    | 0.4545994  |
| GCW  | WC    | 0.0280      | 2.9479      | 0.7256077    | 0.6498312  |
| GCW  | SW    | 0.0139      | 2.5773      | 0.2167222    | 0.3922547  |
| MSM  | UP    | 0.0000      | 2.7503      | 0.6668660    | 0.6377540  |
| MSM  | TR    | 0.0320      | 3.4657      | 1.5193636    | 0.8558215  |
| MSM  | WC    | 0.0000      | 2.9515      | 0.7898891    | 0.6440407  |
| MSM  | SW    | 0.0485      | 0.6284      | 0.2250595    | 0.1193798  |
| GWI  | UP    | 0.0000      | 0.8919      | 0.2869509    | 0.2057429  |
| GWI  | TR    | 0.0187      | 1.7007      | 0.4825754    | 0.3171724  |
| GWI  | WC    | 0.0136      | 1.3526      | 0.5717290    | 0.2670010  |
| GWI  | SW    | 0.1093      | 0.7375      | 0.3690400    | 0.1299047  |

Table 3: PO4 Summary Statistics

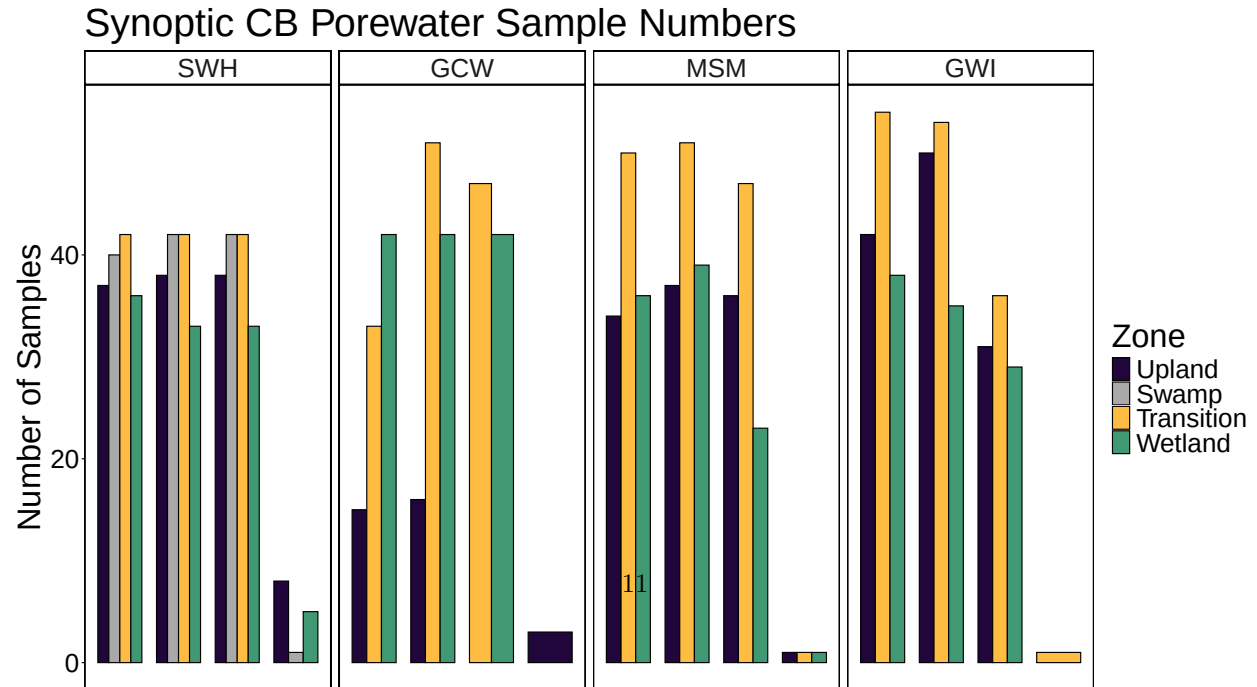
| Site | Zone  | min_PO4_mgL | max_PO4_mgL | mean_PO4_mgL | sd_PO4_mgL |
|------|-------|-------------|-------------|--------------|------------|
| SWH  | UP    | 0.0025      | 0.0800      | 0.0099500    | 0.0125863  |
| SWH  | SWAMP | 0.0027      | 0.4616      | 0.0449425    | 0.0862937  |
| SWH  | TR    | 0.0048      | 7.3861      | 0.5570234    | 1.3267680  |
| SWH  | WC    | 0.0072      | 0.5327      | 0.1075750    | 0.1164197  |
| SWH  | SW    | 0.0057      | 0.0246      | 0.0144694    | 0.0052343  |
| GCW  | UP    | 0.0007      | 1.1614      | 0.0848323    | 0.2605158  |
| GCW  | TR    | 0.0000      | 0.0678      | 0.0121798    | 0.0115546  |
| GCW  | WC    | 0.0000      | 0.8173      | 0.1861188    | 0.1280640  |
| GCW  | SW    | 0.0000      | 0.6966      | 0.0740533    | 0.1056511  |
| MSM  | UP    | 0.0000      | 0.1752      | 0.0180859    | 0.0320906  |
| MSM  | TR    | 0.0000      | 0.7508      | 0.0223736    | 0.0653783  |
| MSM  | WC    | 0.0035      | 0.0920      | 0.0369739    | 0.0240119  |
| MSM  | SW    | 0.0007      | 1.2569      | 0.0336524    | 0.1933594  |
| GWI  | UP    | 0.0000      | 0.1907      | 0.0322407    | 0.0356289  |
| GWI  | TR    | 0.0000      | 0.1039      | 0.0132993    | 0.0161454  |
| GWI  | WC    | 0.0024      | 0.0534      | 0.0093634    | 0.0094964  |
| GWI  | SW    | 0.0000      | 0.0261      | 0.0106178    | 0.0065775  |

Table 4: Total Samples: 1677

| Year | Number_of_samples |
|------|-------------------|
| 2022 | 314               |
| 2023 | 820               |
| 2024 | 522               |
| NA   | 21                |

## 0.9 Summary Tables

### 0.10 Number of Samples



## 0.11 Write out summarized data